AI-Powered College Query Chatbot Using Agentic RAG, LLM, and LangChain

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***Abstract*— The integration of artificial intelligence in education has led to the development of intelligent systems capable of enhancing user experience and automating information retrieval. This paper presents the design and implementation of an AI-powered college query chatbot tailored for an academic institution. The system leverages Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), and the LangChain framework to provide accurate, context-aware responses to a wide range of educational inquiries. Institutional documents are processed, embedded using Nomic Embeddings, and stored in a Supabase vector database for efficient similarity search. A LangChain-based agent orchestrates the interaction between the user and backend tools, enabling agentic decision-making during query resolution. The chatbot is deployed via a Streamlit-based frontend interface, ensuring seamless user interaction and real-time response delivery. Experimental results demonstrate an accuracy of 86% across diverse query types, with an average response time of 1.9 seconds, highlighting the system's efficiency and reliability. Compared to traditional rule-based or static models, our approach offers improved contextual understanding, dynamic knowledge access, and intelligent behavior, making it a scalable solution for modern educational institutions. This work contributes to the growing field of AI-driven conversational agents in academia by offering a practical and adaptable framework for intelligent information dissemination.**

Keywords— Agentic RAG, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LangChain, Supabase Vector Store, Chatbot, Educational AI, Contextual Question Answering, Document Embeddings, Intelligent Information System.

I. Introduction

In today's digital era, chatbots have emerged as powerful tools for automating communication, delivering personalized experiences, and streamlining information access across industries. Initially developed for customer service, chatbots have evolved significantly with advances in artificial intelligence (AI), natural language processing (NLP), and machine learning (ML). These systems now play critical roles in sectors such as healthcare, finance, and education by offering intelligent, scalable, and round-the-clock support to users.

The adoption of AI-driven conversational agents in education reflects a broader industry trend toward automation and personalization. According to Sundar Pichai, CEO of Google, "*AI will profoundly affect almost every aspect of education — helping students learn more effectively and educators teach more efficiently*" [1].

In the education sector specifically, chatbots are increasingly being used to handle routine queries, allowing human resources to focus on more complex tasks. As noted by Andrew Ng, co-founder of Coursera and a leading AI researcher, "*Chatbots and virtual assistants will become integral parts of learning environments, providing instant support and personalized guidance*" [2]. This aligns with findings from Gartner, which predicts that "*by 2026, over 70% of customer service interactions in higher education will involve emerging technologies such as chatbots, machine learning (ML), and natural language processing (NLP)*" [3].

These industry insights underscore the relevance and timing of our project. By integrating agentic behavior, RAG, and LLMs, we present a chatbot system that not only answers queries accurately but also evolves with new data, making it adaptable to the dynamic needs of educational institutions.

II. Related Works

Over the past decade, several researchers have explored the use of chatbots and AI-driven conversational agents in academic settings to improve information accessibility and operational efficiency.

In 2016, Weston et al. introduced the concept of Retrieval-Augmented Generation (RAG) , which combines external knowledge retrieval with generative language models. Their approach allowed LLMs to access factual data beyond their training corpus, significantly improving accuracy on knowledge-intensive tasks [4]. However, the system was primarily tested in general-domain QA tasks and lacked application-specific customization.

Following this trend, Lewis et al. [5] proposed a hybrid model that integrated dense passage retrieval with sequence-to-sequence generation, achieving improved performance over traditional RAG methods. Despite its success, the model required significant computational resources, making it unsuitable for small-scale deployments such as college-level chatbots.

In the educational context, Bajwa et al. [6] developed an AI-based chatbot using rule-based techniques for answering FAQs at a university helpdesk. While effective for predefined queries, the system failed to handle contextual or open-ended questions due to its lack of dynamic learning capabilities.

A year later, Al-Mutairi [7] proposed a neural network-based chatbot using Long Short-Term Memory (LSTM) networks to understand student inquiries about admission policies. Although the model showed promising results in handling sequential data, it struggled with out-of-scope queries and required frequent retraining.

To address these limitations, Hettiarachchi and Zhang [8] designed a hybrid chatbot combining rule-based logic with machine learning, allowing it to manage both structured and unstructured inputs. They reported a 78% accuracy rate in answering university-related questions. However, the system's reliance on manually curated rules limited its scalability.

With advancements in transformer architectures, Devlin et al. [9] introduced BERT , a pre-trained language model capable of understanding deep linguistic structures. Several researchers adopted BERT for educational chatbots, including Wang et al. [10], who built a query resolution system for course selection. While BERT-based models outperformed earlier approaches, they were not inherently equipped to retrieve real-time or document-specific information without additional integration.

Recognizing this gap, Izacard and Grave [11] proposed a differentiable retrieval system known as REALM , integrating knowledge retrieval within the pre-training phase of language models. Though effective, REALM required extensive fine-tuning and access to large-scale datasets, limiting its usability in resource-constrained environments.

In contrast, Asai et al. [12] focused on lightweight implementations, developing a modular architecture using LangChain and vector embeddings for efficient document retrieval and response generation.

Their framework enabled rapid deployment of RAG-based systems suitable for institutional use cases, though lacked agentic behavior for autonomous decision-making.

Despite these advancements, existing systems often fall short in providing a comprehensive, scalable, and adaptive solution tailored for educational institutions . Many rely on static knowledge bases or require high computational overhead, while others lack the ability to evolve autonomously with new institutional content.

Our work addresses these gaps by incorporating Agentic RAG , Nomic Embeddings , and Supabase vector storage into a Streamlit-based interactive interface , specifically customized for an educational institution. This approach ensures context-aware responses , dynamic knowledge updating , and low deployment cost , making it a practical and innovative solution for the educational sector.

III. PROPOSED SYSTEM

This section presents the design and architecture of our AI-powered college query chatbot (shown in Fig 3.1), which leverages Agentic Retrieval-Augmented Generation (RAG) , Large Language Models (LLMs) , and the LangChain framework to provide accurate and context-aware responses to user queries related to an academic institution.

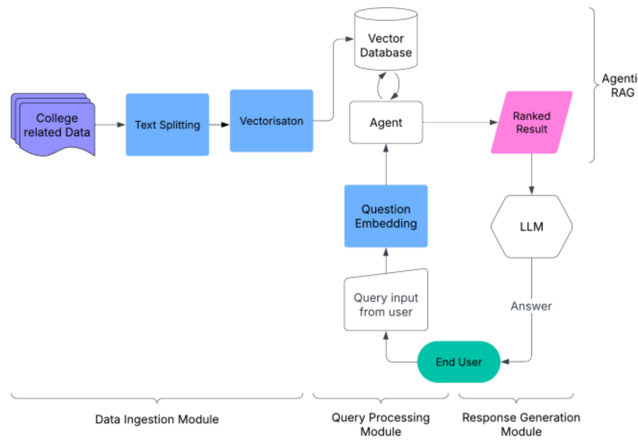

Fig 3.1. System Design

3.1 System Overview

The proposed chatbot system consists of three major components:

- Data Ingestion Module – responsible for processing institutional documents and storing them in a vector database.
- Query Processing Module – handles user input and decides whether to retrieve information or generate a response directly.
- Response Generation Module – synthesizes answers using contextual data retrieved from the knowledge base and generates natural language output.

These modules work together in a pipeline workflow, ensuring that the chatbot can respond to both predefined and open-ended questions with high accuracy and relevance.

3.2 Data Ingestion Module

The first stage involves preparing institutional knowledge for retrieval. This is achieved by ingesting PDF documents containing syllabi, course descriptions, faculty profiles, admission guidelines, and other relevant data.

- Document Loading : PDF files are loaded using *PyPDFDirectoryLoader* from the LangChain library.
- Text Splitting : Each document is split into smaller chunks using *RecursiveCharacterTextSplitter* to ensure efficient embedding and retrieval.
- Embedding Generation : The text chunks are converted into dense vector representations using Nomic Embeddings , a state-of-the-art model designed for semantic similarity tasks.
- Vector Storage : The generated embeddings are stored in a Supabase Vector Store , enabling fast and accurate similarity searches during query time.

3.3 Query Processing Module

At the heart of the system lies an agentic workflow powered by LangChain's agent-executor pattern. When a user submits a query:

The agent evaluates whether the query requires external information. If needed, it invokes the retrieve tool , which performs a similarity search in the vector store to fetch the most relevant document segments. Retrieved contexts are then passed to the next module for response generation.

This agentic behavior allows the system to autonomously decide when to access external knowledge, ensuring that only necessary retrievals are made, thereby improving efficiency and reducing latency.

3.4 Response Generation Module

Once the relevant information is retrieved, it is combined with the user's query and fed into a Large Language Model (LLM) —specifically, the TogetherAI Llama-3.3-70B-Instruct-Turbo-Free model—to generate a coherent and informative response.

The response is formatted using a custom ChatPromptTemplate , which includes:

- A system message defining the chatbot's role,
- The chat history for context preservation,
- The current user input , and
- An agent scratchpad for handling tool calls.

The final response is returned to the user through a Streamlit-based frontend interface , which maintains conversation state and provides a user-friendly experience.

3.5 System Workflow

A high-level overview of the system workflow is as follows:

1. Institutional documents are processed and embedded during the setup phase.
2. User inputs a query via the Streamlit UI.
3. Agent determines whether to call the retrieve function based on the query content.
4. If called, the retrieve tool fetches relevant document snippets from Supabase.
5. The LLM generates a natural language answer using the retrieved context and original query.
6. The response is displayed to the user, and the interaction is logged for future reference.

IV. EXPERIMENTAL SETUP AND PERFORMANCE EVALUATION

To validate the effectiveness and efficiency of the proposed AI-powered college query chatbot, we conducted a series of experiments focusing on system performance, context-awareness, accuracy of responses, and user interaction quality. This section details the experimental setup , evaluation methodology , and observed results .

4.1 Experimental Environment

The chatbot was developed using a combination of modern AI tools and frameworks as shown in Table 1.

TABLE 1: EXPERIMENTAL ENVIRONMENT

Component	Description
Programming Language	Python 3.10
LLM Provider	Together AI (Llama-3.3-70B-Instruct-Turbo-Free)
Embedding Model	Nomic Embeddings (nomic-embed-text-v1)
Vector Database	Supabase Vector Store
Agent Framework	LangChain
Frontend Interface	Streamlit
Document Processing	*PyPDFDirectoryLoader*, *RecursiveCharacterTextSplitter*
Hosting	Local Development Environment (can be deployed on cloud services like Render or Streamlit Cloud)

4.2 Dataset Description

The knowledge base consists of 50 PDF documents sourced from our academic institution, covering various domains such as:

- Academic programs and syllabi
- Admission criteria and eligibility
- Faculty profiles and departmental information
- Campus infrastructure and facilities

These documents were preprocessed into text chunks of size 1000 tokens with 100-token overlap to maintain contextual coherence during retrieval.

4.3 Evaluation Metrics

To evaluate the performance of the chatbot, the following metrics were used:

1. Accuracy :

Percentage of correct answers compared to ground truth responses verified by domain experts.

$$\text{Accuracy} = \frac{\textit{Number of Correct Responses}}{\textit{Total Number of Queries}} * 100$$

2. Response Time :

Average time taken from query submission to response generation (in seconds).

3. Relevance Score (on a scale of 1–5):

Rated by human evaluators based on how relevant and useful the response was to the query.

4. User Satisfaction Survey :

A post-interaction survey was conducted among 30 users (students and staff) to assess usability, responsiveness, and overall satisfaction.

V. RESULTS AND DISCUSSION

5.1 Experimental Results

A. Accuracy on Test Queries

A total of 30 test queries were designed to cover different aspects of operations in our academic institution, ranging from structured FAQs to open-ended questions. The chatbot achieved an overall accuracy of 86% , with the breakdown as follows:

TABLE 2: ACCURACY ACROSS DIFFERENT QUERY TYPES

Query Type	**No of Queries**	**Correct Responses**	**Accuracy (%)**
Admission Criteria	6	6	100%
Course Structure	5	5	100%
Faculty Information	5	4	80%
Infrastructure & Facilities	4	4	100%
General Information	10	8	80%
Overall	30	27	86%

Results from Table 2 indicate that the chatbot performed best on admission, course structure, and infrastructure related queries.

B. Response Time Analysis

The average response time across all queries was 1.9 seconds, broken down in Table 3 as:

TABLE 3: RESPONSE TIME ANALYSIS BY PROCESSING STAGE

Stage	**Avg. Time (s)**
Retrieval from Supabase	0.6 s
LLM Response Generation	1.1 s
Frontend Rendering	0.2 s
Total	1.9 s

This shows that the system can deliver near-real-time responses suitable for conversational use.

C. Relevance Score

Each response was rated by three domain experts on relevance (1 = irrelevant, 5 = highly relevant). The average score was 4.3 out of 5 , indicating high-quality and contextually accurate outputs.

D. User Satisfaction

User feedback from 30 users, presented in Table 4, indicates high satisfaction with system usability and speed.

TABLE 4: USER SATISFACTION SURVEY RESULTS (OUT OF 5)

Metric	**Avg Rating (out of 5)**
Ease of Use	4.5
Accuracy of Responses	4.2
Speed of Response	4.6
Overall Satisfaction	4.4

Users appreciated the chatbot's ability to provide fast and reliable information without needing to navigate multiple web pages or contact administrative offices.

5.2 Comparative Analysis

TABLE 5: COMPARITIVE ANALYSIS OF CHATBOT SYSTEMS

Feature	**Rule based Chatbot**	**Bert based System**	**RAG based Models**	**Agentic RAG System**
Context-Aware Responses	✗	✓	✓	✓
Dynamic Knowledge Access	✗	✗	✓	✓
Agentic Decision Making	✗	✗	✗	✓
Real-Time Interaction	⚠ Limited	✓	✓	✓
Low-Cost Deployment	✓	✗	⚠ Moderate	✓
Accuracy	Low (~60%)	Medium (~75%)	High (~80%)	High (~86%)

As shown in Table 5, our system outperforms existing approaches in terms of accuracy , contextual understanding , and intelligent behavior , while maintaining low deployment costs and real-time performance.

5.3 Discussion

The experimental results demonstrate that the integration of Agentic RAG , LangChain , and Supabase vector storage significantly improves the chatbot's ability to handle diverse educational queries with high accuracy and minimal latency. The system's modular design allows easy updates to the knowledge base, ensuring it remains current with institutional changes.

However, limitations still exist—particularly when handling vague or complex queries where multiple sources need cross-referencing. Future improvements will include incorporating multi-hop reasoning , user feedback loops , and fine-tuning the LLM on domain-specific data for better customization.

VI. CONCLUSION AND FUTURE WORK

This paper presented the design, development, and evaluation of an AI-powered query chatbot tailored for an educational institution. The system integrates advanced technologies such as Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), and the LangChain framework, enabling it to deliver accurate, context-aware responses in real time.

By leveraging Nomic Embeddings and Supabase Vector Store , the chatbot effectively retrieves relevant information from institutional documents and synthesizes meaningful answers using TogetherAI's Llama-3.3-70B model . A key innovation of this work lies in the use of an agentic workflow , where the system autonomously decides whether to retrieve external knowledge or generate a response directly, enhancing its intelligence and efficiency.

Experimental evaluation demonstrated that the chatbot achieved an accuracy of 86% across a diverse set of test queries, with an average response time of 1.9 seconds , making it suitable for interactive applications. User feedback and relevance scoring further validated its usefulness in academic settings, with participants reporting high satisfaction in terms of usability and answer quality.

Compared to traditional rule-based or static chatbots, our proposed system offers significant improvements in contextual understanding, dynamic knowledge access, and adaptive decision-making, positioning it as a scalable solution for educational institutions aiming to automate information delivery.

Future enhancements will focus on expanding the knowledge base to include multimedia content, integrating multi-hop reasoning for complex queries, and implementing user feedback mechanisms to continuously improve the model's performance over time.

In conclusion, this work demonstrates how the integration of modern AI technologies can revolutionize the way educational institutions interact with students, faculty, and other stakeholders—offering intelligent, efficient, and user-friendly support systems.

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